MILITARY SPECIFICATION

REGULATOR, PRESSURE

This specification is approved for use within the Army Missile Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements for two types of regulating assemblies for control and indication of discharge pressure from storage cylinders.

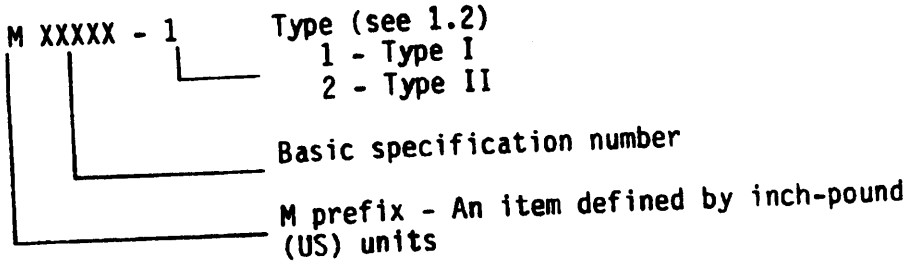
1.2 Classification. The regulating assemblies shall be of the following types:

TYPES

Type I - 2500 pounds per square inch (psi), maximum outlet pressure
Type II - 1000 pounds per square inch (psi), maximum outlet pressure

1.3 Part number. The part number shall consist of the letter M, the basic number of this specification and dash numbers denoting type.

Example:



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, Al, 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

FSC 4820

AMSC N/A

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)

MILITARY

MIL-P-7105	Pipe Threads, Taper, Aeronautical, National Form, Symbol ANPT, General Requirements For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys

STANDARDS

FEDERAL

FED-STD-H28/9	Screw-Thread Standards for Federal Services Section 9 Gas Cylinder Valve Outlet and Inlet Threads

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging of
MIL-STD-889	Dissimilar Metals

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MS33656 Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of a good commercial quality suitable for the purpose. Materials shall be free from all defects and imperfections that may affect the serviceability of the finished product. Dissimilar metals as defined by MIL-STD-889 shall not be placed in contact with each other unless the adjoining surfaces are suitably coated, sealed, or applied with an interposing material to preclude any corrosive action.

3.2.1 Nonmetallic material. Nonmetallic material that deteriorates or is otherwise affected adversely by continued use with high pressure water pumped nitrogen shall not be used.

3.2.2 Nonferrous materials. Nonferrous materials shall be used in the construction of the regulator, except where ferrous materials are essential.

3.2.3 Protective treatment. Construction materials, subject to deterioration when exposed to climatic and environmental conditions during service usage, shall be protected in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip or scale with age or extreme climatic and environmental conditions shall be avoided.

3.2.4 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in the order of precedence of MIL-STD-143.

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3.3 Design and construction.

3.3.1 Regulator. The regulator shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand strains, jars, vibrations, and other conditions incident to shipping, installation, and service. The regulator shall neither give off nor contain any oil or flammable compound that would contaminate water pumped nitrogen passing through it to various equipment. The regulator shall be so constructed that adjustments and repairs can easily be made without the use of special tools.

3.3.2 Regulator operation. The regulator shall be designed and constructed to reduce pressure from an initial source to a required delivery pressure. The pressure adjustment control shall consist of a thread screw provided with a knob type handle which will require a maximum of 30 inch-pounds of torque to turn. The final delivery pressure shall not vary more than three percent from the preset value. When the pressure adjustment control is turned counter-clockwise, the outlet pressure of the regulator shall be vented (see 4.6.2).

3.3.2.1 Regulator (type I). The regulator shall reduce pressure from a source up to 3000 psi to a delivery pressure between 0 to 2500 psi and flow up to 50 standard cubic feet per minute (SCFM) (see 4.6.2).

3.3.2.2 Regulator (type II). The regulator shall reduce pressure from a source up to 3000 psi to a delivery pressure between 0 to 1000 psi and flow up to 50 SCFM (see 4.6.2).

3.3.3 Inlet screen. A filter, equivalent to a 200-mesh screen, shall be provided in the inlet connection to prevent foreign particles from entering the regulator and shall be removable for cleaning (see 4.6.1.1).

3.3.4 Pressure gages. The regulator shall be provided with two commercial gages, 2 or 2.5 inches in diameter. The accuracy for each gage shall be within the commercial tolerances established for this type of equipment, except the tolerance over the range between 33.33 and 66.66 percent of the maximum scale reading shall not exceed 2 percent of the maximum reading to which the scale is graduated. The error in reading at any point on the scale shall not exceed 3 percent of the maximum reading to which the scale is graduated (see 4.6.1.1).

3.3.4.1 Regulator (type I) pressure gages. The inlet pressure gage shall indicate pressures from 0 to 3000 psi; the outlet pressure gage shall indicate delivery pressures from 0 to 3000 psi (see 4.6.1.1).

3.3.4.2 Regulator (type II) pressure gages. The inlet pressure gage shall indicate pressures from 0 to 3000 psi; the outlet pressure gage shall indicate delivery pressures from 0 to 1500 psi (see 4.6.1.1).

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3.3.5 Relief valve. A resealing type relief valve shall be provided in the delivery side of the regulator to prevent excessive build-up of pressure in this chamber. The relief valve shall be preset and sealed to prevent alteration of venting pressure (see 4.6.3).

3.3.5.1 Venting (type I regulator). Type I regulator shall begin at a delivery pressure of 2600 psi and the relief valve shall seal at not less than 2400 psi (see 4.6.3).

3.3.5.2 Venting (type II regulator). Type II regulator shall begin at a delivery pressure in excess of 1200 psi and shall seal at not less than 1000 psi (see 4.6.3).

3.3.6 Screw threads. Unless otherwise specified, all machine screw threads shall conform to MIL-S-7742, class 2.

3.3.7 Pipe threads. Pipe threads shall conform to MIL-P-7105.

3.3.8 Regulator connections (type I and type II). The inlet connection shall be a No. 580 nipple with a No. 580 hex nut in accordance with Connection No. 580 of FED-STD-H28/9. The outlet connection shall be MS33656G4.

3.3.9 Dimensions (type I regulator). The overall dimensions of the regulator shall be not greater than 10 inches by 10 inches by 10 inches, including gages (see 4.6.1).

3.3.10 Dimensions (type II regulator). The overall dimensions of the regulator shall be not greater than 9 inches by 9 inches by 8 inches, including gages (see 4.6.1).

3.3.11 Weight. The weight of the regulator with gages shall not exceed 10 pounds (see 4.6.1).

3.3.12 Finishes and protective coatings. All protected metal parts and surfaces that are constructed from corrosion resistance material and on which a protective coating would be detrimental to proper functioning shall not be given a protective coating. All exposed metal parts and surfaces and all protected metal parts on which a protective coating will not be detrimental to proper functioning shall be treated to satisfy the requirements of this specification.

3.3.13 Magnesium alloy parts. Magnesium base alloy parts shall not be used.

3.3.14 Aluminum alloy parts. Aluminum alloy surfaces shall be anodized in accordance with MIL-A-8625.

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3.3.15 Ferrous parts. All ferrous parts shall be cadmium plated in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.4 Performance characteristics.

3.4.1 Operational environment (type I). The regulator shall be capable of operating satisfactorily under the following conditions (see 4.6.4.1):

a. Temperature of room, 22° Celsius (C) to 24°C.

b. Temperature of gaseous nitrogen supply, 0°C to 50°C.

c. Relative humidity, 20 percent to 50 percent.

d. Altitude from 0 to 10,000 feet (ft) above sea level.

3.4.2 Operational environment (type II). The regulator shall be capable of operating satisfactorily under the following conditions (see 4.6.4.2):

a. Temperature of van, 15.5°C to 30°C.

b. Temperature of gaseous nitrogen supply, 10°C to 32.2°C.

c. Relative humidity, 0 percent to 90 percent.

d. Altitude from 0 to 10,000 ft above sea level.

3.4.3 Nonoperational environment. The regulator shall operate satisfactorily when returned to an operational environment after being subjected to the following conditions (see 4.6.5):

a. Stored for 48 hours at -55°C.

b. Stored for 48 hours at +60°C.

c. Stored for 48 hours at 40,000 ft altitude.

d. Stored for 48 hours at relative humidity of 100 percent.

e. Stored for 48 hours at a relative humidity of 0 percent.

3.4.4 Vibration. The unit shall operate satisfactorily after being subjected to vibration with the frequency varying logarithmically between

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5 to 500 hertz at the following levels (see 4.6.6).

a. 5 to 27 hertz, 1.3 gravity units (g) ± 0.05 g.

b. 27 to 52 hertz, 0.036 ± 0.001 inches double amplitude.

c. 52 to 500 hertz, 5.0 g ± 0.05 g.

3.4.5 Vertical and lateral shock. The regulator shall be capable of operating satisfactorily after being subjected to a vertical and lateral shock of 15 g half-sine pulse with a duration of 11 milliseconds (see 4.6.7).

3.4.6 Tilt drop. The regulator shall be capable of operating satisfactorily without permanent damage or degradation of accuracy after being subjected to a tilt drop test on each of its ends, using the opposite end as a pivot, for a distance of 4.0 inches above a horizontal plane with a free drop to a solid surface (see 4.6.8).

3.4.7 Service life. The type I regulator shall be capable of being cycled 10,000 times between 100 and 2000 psi when connected to a 2200 psi pressure source. The type II regulator shall be capable of being cycled 10,000 times between 0 and 1000 psi when connected to a 2200 psi pressure source. The regulators shall not leak after cycling (see 4.6.9).

3.5 Operational characteristics. The regulator specified herein shall operate satisfactorily in any position in which it is mounted. It shall not hum or vibrate within its operating range of pressure or flow (see 4.6.9).

3.6 Identification. The unit shall be identified in accordance with MIL-STD-130.

3.7 Workmanship. The regulator, including all parts and accessories, shall be fabricated and finished in accordance with high grade commercial practices. Particular attention shall be given to freedom from blemishes, defects, burrs, sharp edges and accuracy of dimensions (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the conditions of 4.3.1 and 4.3.2 of this specification.

4.3.1 Cleaning. Prior to inspection, the regulator shall be thoroughly cleaned and kept free of oil or other flammable compounds (see 5.1.1).

4.3.2 Room temperature and pressure. Unless otherwise specified, all inspections and tests shall be conducted with the regulator at room temperature between 15.5°C to 30°C and at atmospheric pressure.

4.4 First article inspection.

4.4.1 First article sample. When specified in the contract or purchase order (see 3.2 and 6.2), the first article sample shall consist of one regulator, representative of the production product. The first article sample shall be subjected to the inspections and tests specified in table I in the order shown. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity.

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TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Examination of product	3.3.10, 3.3.11, 3.3.12, 3.6 and 3.7	4.6.1
Regulator pressure and flow	3.3.2, 3.3.2.1 and 3.3.2.2	4.6.2
Relief valve calibration	3.3.5	4.6.3
Operational environment	3.4.1 and 3.4.2	4.6.4.1 & 4.6.4.2
Nonoperational environment	3.4.3	4.6.5
Vibration	3.4.4	4.6.6
Shock	3.4.5	4.6.7
Tilt drop	3.4.6	4.6.8
Service life	3.4.7 and 3.5	4.6.9

4.5 Quality conformance inspection. The quality conformance inspection shall consist of individual inspection and test on each regulator.

4.5.1 Inspection and tests. Each regulator shall be subjected to the inspection and tests specified in table II in the order shown.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Examination of product	3.3.10, 3.3.11, 3.3.12, 3.6 and 3.7	4.6.1
Regulator pressure and flow	3.3.2, 3.3.2.1 and 3.3.2.2	4.6.2
Relief valve	3.3.5	4.6.3

4.5.2 Noncompliance. If a regulator fails to pass inspection it may be reworked to correct defects and resubmitted for inspection. Before reinspection, full particulars concerning the noncompliance and the action taken to correct the defects shall be furnished to the procuring activity. Acceptance and shipment shall be discontinued until corrective action acceptable to the procuring activity has been taken. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the procuring activity.

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4.5.3 Inspection of packaging. Preservation, packing and marking shall be inspected for conformance to section 5.

4.6 Methods of inspection.

4.6.1 Examination of product. Each regulator shall be visually and dimensionally inspected and weighed to determine conformance with the requirements of this inspection (see 3.3.10, 3.3.11, 3.3.12, 3.6 and 3.7).

4.6.1.1 Certification of requirements. The manufacturer shall certify that the requirements of 3.3.3 and 3.3.4 are in accordance with this specification.

4.6.2 Regulator pressure and flow (type I and II). The regulator inlet shall be pressurized to 3000 psi and the regulator tested for conformance to 3.3.2.

4.6.3 Relief valve (type I and II). The regulator inlet shall be pressurized to 3000 psi and the regulator tested for conformance to 3.3.5.

4.6.4 Operational environment.

4.6.4.1 Type I. The regulator shall be stabilized at the conditions of 3.4.1 and operated in accordance with 3.3.2 and 3.3.5.

4.6.4.2 Type II. The regulator shall be stabilized at the conditions of 3.4.2 and operated in accordance with 3.3.2 and 3.3.5.

4.6.5 Nonoperational environment (type I and II). The regulator shall be stored at the conditions of 3.4.3, returned to room temperature and operated in accordance with 3.3.2 and 3.3.5.

4.6.6 Vibration (type I and II). The regulator shall be subjected to the vibration levels of 3.4.4 and then operated in accordance with 3.3.2 and 3.3.5.

4.6.7 Shock (type I and II). The regulator shall be subjected to the shock levels of 3.4.5 and then operated in accordance with 3.3.2 and 3.3.5.

4.6.8 Tilt drop. The regulator shall be tested for conformance to 3.4.6.

4.6.9 Service life (type I and II). The regulator inlet shall be pressurized to 2200 psi and cycled in accordance with the requirements of 3.4.7. There shall be no leakage at the end of 10,000 cycles.

5. PACKAGING

5.1 Preservation. Preservation shall be level C of MIL-STD-794.

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5.1.1 Cleaning and drying. Cleaning and drying shall be in accordance with MIL-STD-794.

5.1.2 Preservation applications. Preservatives shall not be used.

5.2 Packing. Packing shall be level C of MIL-STD-794.

5.3 Marking.

5.3.1 Level C. In addition to any special or other identification marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I regulator. The type I regulator is intended for use in a water pumped nitrogen system in a laboratory.

6.1.2 Type II regulator. The type II regulator is intended for use in a water pumped nitrogen system in a mobile van.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification.

b. Part number (see 1.3).

c. Whether first article sample is required (see 3.1 and 4.4).

d. Special markings required (see 3.6 and 5.3).

e. Whether certification of compliance is required (see 4.6.1.1).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and

gaseous nitrogen supply: 0°C to 50°C.

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that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Regulator, pressure
Valve, regulator

6.5 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

Custodian:
Army - MI

Preparing activity:
Army - MI

Review Activities:
DLA - CS

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-47321(MI)	REGULATOR, PRESSURE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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